BESS VS DIESEL BACKUP

It is a runtime question, not a technology one

24 hours of battery backup costs about 8x what diesel does per MW

THE DATA

What backup really costs

$0.8–1.2M/MW

Tier 4 diesel (Schneider Electric WP14)

$7.0–9.5M/MW

24-hour BESS (Schneider Electric WP14)

Rs 25–40/kWh

diesel power cost in India (Power Line, Aug 2026)

ROOT CAUSE

Why the pressure is building

- US: emergency gensets get 100 h/yr non-emergency use, only 50 beyond testing
- Virginia: new data-centre permits after 1 Jul 2026 must meet Tier 4-level limits
- India: CPCB IV+ lifted genset prices 20–40%
- Tier 4 retrofit: $100k–500k+ per engine

WORKED EXAMPLE (ILLUSTRATIVE)

10 MW hall, priced by runtime

~$0.8–1M

15-minute bridge, 2.5 MWh

~$12–16M

4-hour ride-through, 40 MWh

~$8–12M

diesel for the same load

Implied ~$0.3–0.4M per MWh from the Schneider range

WHAT TO DO

Design it hybrid

- Split the outage curve: price minutes, hours and days separately
- Let BESS earn via demand response or peak shaving
- Write SLA, fire-safety and augmentation risk into the contract
- Ask for UL 9540A data and NFPA 855 siting up front
- Model diesel with a carbon or permit-cost scenario

SOURCES

Data behind this carousel

- Schneider Electric White Paper 14: BESS vs diesel backup
- Power Line, India genset market (6 Aug 2026)
- Facilities Dive, demand response risk
- AJOT and Virginia Mercury on backup-power emissions

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